

# Samhitha Kondeti

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## Education

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### Texas A&M University, College Station, TX

Aug 2025 - May 2027

Master of Science in Computer Science (GPA: 4.0/4.0)

- Coursework : Natural Language Processing, Cybersecurity Risk, Analysis of Algorithms, Intelligent User Interfaces, Quantum Algorithms, Computers and New Media.

### National Institute of Technology, Durgapur

Dec 2020 - May 2024

B.Tech in Computer Science and Engineering (GPA: 8.65/10)

- Coursework : Computer Networks, OS, Software Engineering, DSA & OOP, Data Analysis, DBMS.

## Skills

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**Programming Languages:** C/C++, Java , Python(NumPy, Pandas), Rust, Html, CSS, JavaScript, SQL, Lean, LaTeX

**ML Frameworks & Tools:** PyTorch, React Native, Jupyter Notebook, Hugging Face, Git/GitHub, AWS, TxGraffiti

**Interests:** University Choir & Performance, Yoga/Wellness, Logic Puzzles (Sudoku) & Japanese

## Experience & Research

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### When LLM Self-Correction Helps vs Hurts

Feb 2026 - April 2026

Lead Research Author, Academic Project

- Conducted a 96 condition empirical study (71,595 pairs, 3 LLMs, 4 benchmarks) proving prompting based self-correction is net harmful (3.4× damage ratio) and that self-consistency outperforms it at equal compute .
- Built two novel SFT datasets with DeepSeek-R1 step attribution and LoRA fine tuning, achieving +17.82 pt GSM8K on Mistral-7B, and the model regresses on code tasks where external verifiers exist (Codeforces 15.22)

### Quantum Circuit Decomposition via Automated Conjectures

March 2026 - Present

Researcher, Advisor: Prof. Andreas Klappenecker, TAMU

- Investigating optimal quantum circuit decomposition of controlled unitary gates using TxGraffiti, an automated conjecture tool, to discover mathematical relationships between circuit structure and graph invariants. .
- Exploring connections between Graph Theory and Quantum circuit complexity through Automated Reasoning; conducting systematic search for conjectures with verified counterexample detection. [TxGraffiti2, Python, Qiskit]

### Comparative Analysis of QGD on a Continuous TSP

Feb 2026 - Present

- Demonstrated via 1,000 Monte Carlo trials that Simulated Quantum Gradient Descent (QGD) outperformed Classical Gradient Descent by 13+ percentage points in finding the global minimum, isolating gradient estimation quality as the critical factor on non-convex landscapes. [Tools: Qiskit, Python, SciPy (BFGS optimizer), LaTeX ]

### GNN for granular visualization of water distribution in cities

Jan 2025 - Jun 2025

Research Assistant, NIT Durgapur

- Developed an advanced deep learning model under Prof. Suchismita Roy by integrating sensor data from over 10 municipal stations, processing 10,000 of daily data points to capture spatial correlations. [PyTorch Geometrics]

## Position of Responsibility

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### Sponsorship Head, ISTE NIT Durgapur

Apr 2023 - May 2024

- Secured \$4,500 in sponsorship funding and grew community LinkedIn following to 4K.

### Class Representative, NIT Durgapur

Apr 2022 - May 2024

- Served as the primary liaison between faculty and a cohort of 190+ students.

## Work Authorization

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- Eligible to work in the U.S. under Curricular Practical Training (CPT) without sponsorship.